STAT 6560
Graphical Methods

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Project Two

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splancs: Spatial and Space-Time Point Pattern Analysis Functions

Spatial Point Processes

Spatial point processes are defined as random point observations in space (or time). The locations of the points are random before they are observed (ie, they are not chosen and fixed as part of the experiment). Measurements made at the random location may also be random.

Examples:

- Locations of trees in a timber harvest stand.
- Locations of craters on the moon.
- Location of lung cancer cases in relation to location of pollution source.

The three main types of point patterns consist of: complete spatial random (CSR), clustered, and regular. Clustered point processes are just as you would imagine: points that are clustered together at certain distances. Regular point processes are points that seem to follow a grid pattern. CSR point patterns consist of points that fall between being clustered and regular. Here are some examples of each type of point processes patterns. These types of data can be analyzed spatially, using the R package splancs.

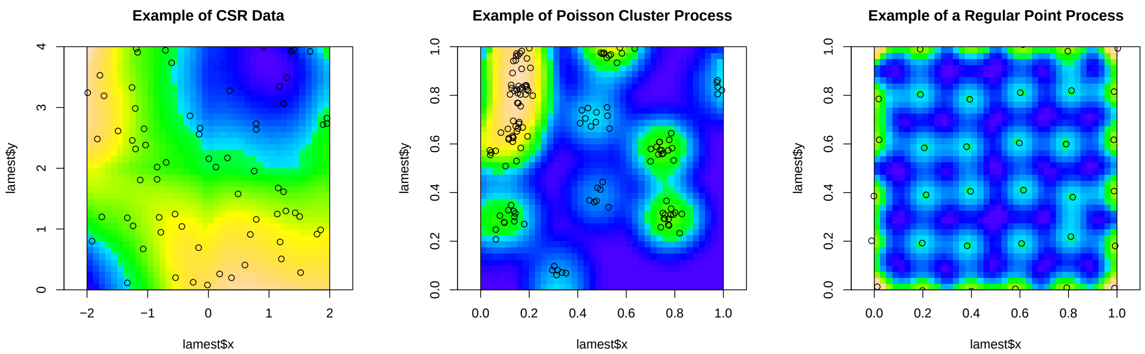

Figure 1: Density plots of a csr process, a poisson cluster process, and a jittered grid.

splancs

Splancs contains many useful functions for the analysis of spatial data. I will focus on the k-function and the l-function. These functions give a graphical representation of whether the data locations are clustered, regular, or complete spatial random (csr). We can estimate k-hat by:

$$\hat{k} = \frac{1}{\lambda^2 |D|} \sum \sum I_h(d_{ijth})$$

where D is the area of the polygon D,
d_{ij} is the distance between the i^{th} and j^{th} observed event in $|D|$,
$\mathrm{I}(\mathrm{d}_{ij}) = 1$, if $d_{ij} \leq h;$ or 0, otherwise.

The k-hat plots can be hard to interpret. So instead we use l-hat defined by:

$$\hat{l}(h) = \sqrt{\frac{\hat{k}(h)}{\pi}} - h$$

The following plots are the l-functions of our previous examples of point processes.

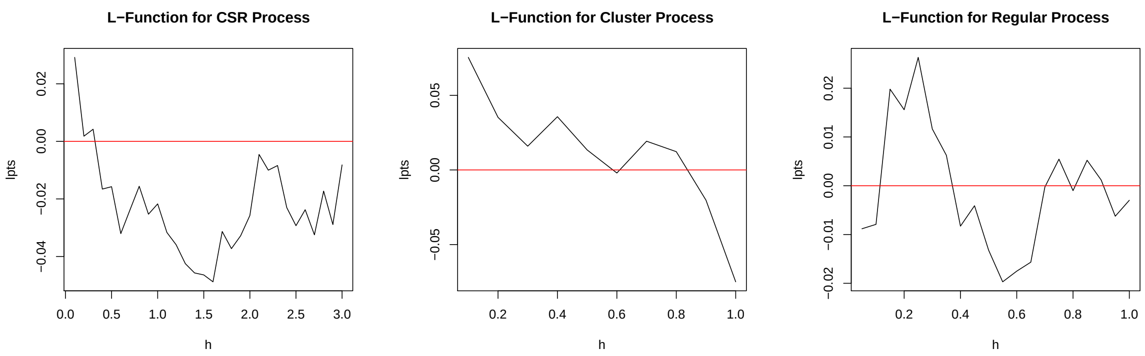

Figure 2: L-functions of a csr process, a poisson cluster process, and a jittered grid.

We can create a sort of confidence interval to show the bounds in which we would expect these l-functions to lie. These plots are shown below.

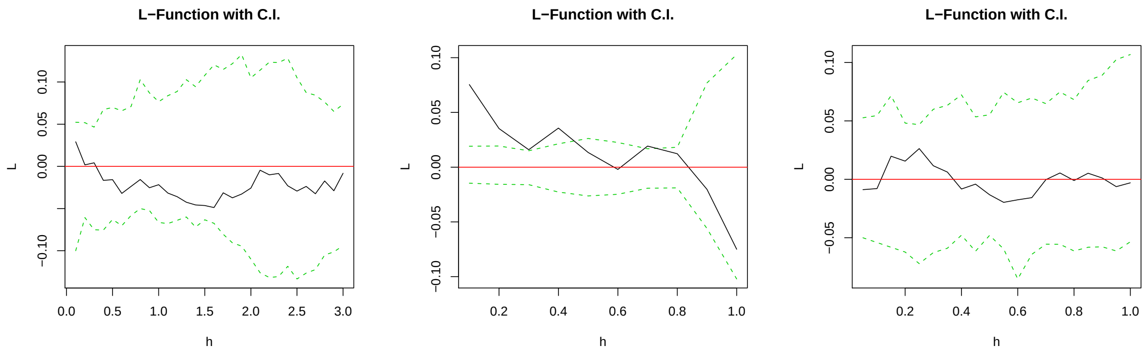

Figure 3: L-functions with confidence intervals of a csr process, a poisson cluster process, and a jittered grid.

Examples in R (R code found below):

- Simulated Homogeneous Poisson Point Process
- Female and Male coyote locations in a region of Texas Hooten et al. (2008)
  - Locations of these animals was of interest to ecologists for a number of reasons. We will examine these data to see where coyotes live in relation to one another. In essence, we are looking for distances at which the coyotes are clustered together or are away from each other in a grid-like fashion.

Links to Data files and R-code:

- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project2_poisson.R`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project2_coyotes.R`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project2_female.txt`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project2_male.txt`

References

Hooten, M. B., Wilson, R. R. & Shivik, J. A. (2008), Hard Core or Soft Core: On the Characterization of Animal Space Use., *in* '2008 JSM Proceedings', American Statistical Association, Alexandria, VA, pp. 1301–1308. (CD).